PANKAJ KUMAR AGARWAL

Principal Backend & Distributed Systems Engineer

Cloud-Native Microservices · Event-Driven Architecture · Node.js · NestJS · TypeScript · Angular (Latest) · Kafka · Kubernetes · Data Lake Architecture

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PROFESSIONAL SUMMARY

Principal Backend & Platform Engineer with 15+ years of experience architecting and delivering cloud-native, microservices-driven platforms using Node.js/NestJS, Golang (Gin, gRPC), and modern frontend frameworks including Angular (latest versions) and React.

Proven track record leading end-to-end architecture and implementation of large-scale enterprise systems across e-commerce, mobility, fintech, healthcare, airlines, and fleet telematics domains. Extensive experience in building distributed systems, event-driven architectures, scalable data platforms, and Data Lake ecosystems on AWS and GCP.

Strong expertise in high-throughput systems, complex business workflows, regulatory-compliant platforms (RBI, AIS-140, healthcare standards), observability, performance optimization, and cost-efficient cloud architecture.

Core Backend

- Node.js / NestJS, Express.js
- REST & GraphQL APIs
- Golang (Gin, Fiber, gRPC)
- Concurrency & parallel processing with goroutines
- Scalable API design and multi-tenant SaaS platforms

Architecture & Patterns

- Microservices Architecture
- Domain-Driven Design (DDD)
- Event-Driven Architecture
- CQRS
- BFF / API Gateway

- Saga / Orchestration Patterns
- Workflow Engines
- Platform Architecture & Multi-tenant System Design
- Data Lake Architecture & Distributed Data Platforms

Messaging & Streaming

- Kafka
- RabbitMQ / other message queues
- AWS SNS / SQS
- GCP Pub/Sub

Datastores & Data Platforms

- PostgreSQL
- MySQL
- MongoDB
- Redis
- GCP: Firestore, Bigtable, BigQuery
- Data Lake implementations using S3 / GCS
- Data ingestion pipelines & event streaming architectures

Cloud Platforms

AWS:
EC2, S3, Lambda, API Gateway, RDS, ECS/EKS, Fargate, CloudWatch

GCP:
GKE, Cloud Run, Cloud Storage, Pub/Sub, BigQuery

Containers & Orchestration

- Docker
- Kubernetes

CI/CD & Automation

- Jenkins
- GitHub Actions

- Infrastructure automation & deployment pipelines

Observability & Monitoring

- Prometheus
- Grafana
- ELK / EFK Stack
- CloudWatch
- OpenTelemetry

API Security

- JWT
- OAuth2
- RBAC & Secure Gateway Patterns

Frontend

- Angular (Latest Versions)
- React.js
- Redux
- React Native
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PROFESSIONAL EXPERIENCE SUMMARY

Ola Electric – Principal Engineer | Dec 2020 – Present

Cisco – UI & Backend Technical Lead | Mar 2020 – Dec

2020 PayPal – Senior Backend Engineer | Apr 2019 – Jan

2020 Flipkart – Principal Software Engineer | Dec 2017 –

Apr 2019 UST Global – Node.js Specialist | Jul 2016 – Oct

2017

Seecog Softwares – Senior Web Developer | Aug 2011 – May 2016

Project Summary

Project 1 :

Ola Electric – End-to-End Electric Scooter Purchase & Delivery Platform

Organization: Ola Electric
Role: Principal Engineer
Duration: Dec 2020 – Present

Overview

Architected and led the development of Ola Electric's end-to-end digital commerce and fulfillment platform enabling customers to complete the entire electric scooter journey — from pre-booking, finance, and test rides to experience centers and last-mile delivery — across web, mobile, and POS channels.

The platform supports multi-role users, high-scale transactional workloads (handling up to 8 lakh users per day during festive peaks), event-driven microservices, real-time tracking systems, and an enterprise-scale Data Lake for analytics and operational intelligence.

Key Responsibilities & Contributions

- Designed and implemented the overall cloud-native microservices architecture covering booking, finance, test rides, experience centers, and last-mile delivery.
- Architected an event-driven system using Kafka for asynchronous communication between services.
- Designed and implemented a centralized Data Lake architecture (Azure Blob + BigQuery/S3-style storage patterns) to ingest transactional, behavioral, and operational events for analytics and BI workloads.
- Built real-time and historical analytics pipelines integrating customer journeys, payments, delivery tracking, and engagement events.
- Integrated and optimized Time-Series Database (TimescaleDB / InfluxDB) for:
    - Delivery state transitions
    - Vehicle movement logs
    - Operational metrics

    - Performance telemetry & monitoring

- Led cross-functional teams across backend, frontend (web/mobile), DevOps, and data engineering.
- Designed secure authentication flows including OTP-based login, MFA (IVR/password), and JWT-based session handling.
- Integrated payment gateways (PayU, UPI, cards, net banking, Ola Money) and handled complex edge cases including partial payments, EMI splits, and refund workflows.
- Implemented dynamic product configuration using Azure-hosted JSON for scooter variants, pricing logic, and feature toggles.
- Architected scalable caching and rate-limiting strategies using Redis to handle festive traffic surges.

Key Functional Modules

🚀 Pre-Booking & Booking (Mobile Web App)

- Variant & color selection
- Dynamic pricing & config-driven journey
- Order booking, modification, cancellation
- High-throughput transactional architecture

💳 Finance & EMI Flow

- Loan eligibility journeys
- EMI plan simulation & selection
- Partial payment handling
- Third-party credit integrations
- Compliance-aligned financial workflows

🛵 Test Ride Platform

- Customer portal with slot booking & Google Maps-based location detection
- Brand Champion React Native app (ride lifecycle management, POS flows)
- Admin dashboard (zone/city logic, champion onboarding, overrides, whitelist/blacklist)
- Feedback SPA & campaign logic
- Real-time slot utilization tracking (time-series storage)

🏬 Experience Center Module

- QR-based onboarding
- Lead capture & geo-redirection
- Tablet/mobile optimized assisted-sales flow

🚚 Last Mile Delivery (LMD) System

- Real-time delivery tracking
- Delivery champion & SDC management
- Address validation & OTP-based fulfillment
- Time-series tracking of delivery lifecycle states

📊 Analytics & Engagement

- Behavioral tracking via MoEngage
- Event streaming into Data Lake
- Centralized analytics for:
    - Conversion funnels
    - Drop-off analysis
    - Delivery SLAs
    - Operational KPIs
- BI-ready structured datasets for leadership dashboards

Technology Stack

Backend

- Node.js microservices (NestJS, Express)
- Event-driven architecture using Kafka
- MongoDB, MySQL (transactional systems)
- Redis (caching, rate limiting)

Data & Analytics

- Data Lake architecture (Azure Blob / Object Storage based ingestion layer)
- BigQuery-style analytics patterns
- Event ingestion pipelines
- TimescaleDB / InfluxDB (time-series storage for tracking & telemetry)

Frontend (Web)

- React.js
- Redux Thunk
- Micro-frontend architecture

Frontend (Mobile)

- React Native

Authentication & Security

- OTP-based authentication
- MFA (Password / IVR)
- JWT-based sessions
- Secure API gateway patterns

Payments

- PayU
- UPI
- Credit/Debit Cards
- Net Banking
- Ola Money

Cloud & Storage

- Azure Blob Storage
- Azure-hosted JSON configuration
- Containerized deployment

Analytics & Monitoring

- MoEngage
- Custom event emitters
- Centralized logging & monitoring
- Real-time operational dashboards

DevOps & Quality

- Dockerized services
- Automated CI/CD pipelines
- SonarQube for code quality

Scale & Impact

- Handled 8 lakh+ users/day during peak festive demand
- Enabled scalable microservices supporting multi-channel commerce

- Improved delivery tracking reliability through time-series data modeling
- Enabled leadership decision-making via centralized Data Lake analytics
- Designed highly resilient, horizontally scalable architecture

Project 2:

Cisco – Enterprise Workforce Management & Expense Automation Platform

Organization: Cisco

Role: Backend Technical Lead

Duration: Mar 2020 – Dec 2020

Overview
Led the design and implementation of an enterprise-grade platform to automate HR and Finance workflows, including expense management, transfers/promotions, deputations, and workforce shift planning. Built for global operations with multi-currency, high concurrency, and complex approval workflows.

Key Responsibilities & Contributions

- Defined architecture for expense management, transfer & promotion workflows, and

shift planning modules.

- Led the full-stack implementation using Golang/Spring Boot microservices with React-based frontends.
- Implemented multi-level workflow approvals, role-based access control, and policy-compliant validation.
- Designed APIs and data models to support multi-currency expense claims and travel request workflows.

Key Functional Modules

- **Expense Management Module:**
  - Submission and validation of business expenses.
  - Multi-currency handling with live forex synchronization.
  - Competence period validations and policy enforcement.
  - Travel request association and workflow approvals.
- **Transfer, Promotion & Deputation Module:**
  - Internal transfers, promotions, and deputations across BUs/geographies.
  - Pre-configured approval hierarchies and document verification.
  - Integration with core HR systems for real-time employee updates.
- **Shift Management Module:**
  - Dynamic definition and assignment of working shifts.
  - Custom working hours per employee/project.
  - Bulk creation of time records for daily/weekly/monthly schedules.

Technology Stack

- **Backend:** Node.js / NestJS microservices (Express.js), REST & GraphQL APIs
- **Middleware:** Node.js with Express.js
- **Frontend:** React.js, ES6, Redux, Redux Thunk
- **APIs & Communication:** GraphQL, REST APIs
- **Event Streaming:** Kafka for asynchronous, event-driven workflows
- **Authentication & Authorization:** JWT, custom RBAC

- **Architecture:** Distributed microservices with high-concurrency support

Project 3:

PayPal – Checkout Lite & Smart Billing Agreement (India Regulatory Compliance)

Organization: PayPal (via employer)
Role: Senior Backend Engineer
Duration: Apr 2019 – Jan 2020

Overview
Contributed to the design and implementation of PayPal's Checkout Lite experience—a lightweight, mobile-optimized checkout flow with smart billing agreement (RT Agreement) logic, tailored to RBI rules for recurring payments in India.

Key Responsibilities & Contributions

- Implemented a responsive, simplified checkout front-end integrated with PayPal's Lite SDK.
- Integrated Billing Agreement / Reference Transaction logic to enable "Link & Pay" repeat payment experiences.
- Implemented India-specific regulatory logic for recurring payments aligned with RBI thresholds.
- Optimized performance focusing on mobile-first usage and minimal resource footprint.

Key Functional Areas

- **Checkout Lite Experience:**
  - Lightweight, responsive checkout for faster load times.
  - Adaptive rendering for various device types and platforms.
- **Smart Billing Agreement (RT):**
  - One-time authorization for seamless repeat payments.
  - Auto-filled payment flows without repeated credential entry.
- **India-Specific Compliance:**
  - CVV bypass for transactions ≤ ₹2000.

- Enforced CVV + 3DS for transactions > ₹2000 per RBI regulations.
- Risk logic integrated with merchant eligibility and transaction limits.

Technology Stack

- **Frontend:** React.js (Lite SDK integration), ES6, responsive web components
- **Backend:** Node.js (middleware), NestJS , mongoDB , postgres
- **APIs:** PayPal REST APIs (Billing Agreements, Reference Transactions)
- **Security & Compliance:** 3DS integration, CVV enforcement rules, RBI-compliant flows
- **Region Logic:** Country and currency-aware service orchestration
- **Performance:** Lazy loading, footprint minimization, mobile UX optimization

Project 4:

Flipkart – Billion System Diagnosis & Fulfillment Track-and-Trace Platform

Organization: Flipkart
Role: Principal Software Engineer
Duration: Dec 2017 – Apr 2019

Overview
Designed and developed internal analytics and fulfillment tools to improve product performance visibility and operational control for Flipkart's private label brand, Billion, and to enhance transparency in the order fulfillment lifecycle.

Key Responsibilities & Contributions

- Led architecture and implementation of the Billion System Diagnosis dashboard.
- Designed the fulfillment Track & Trace service, including action eligibility rules and data storage.
- Defined versioned service abstraction to evolve architecture from tightly coupled to more modular design.

Key Functional Modules

- **Billion System Diagnosis Dashboard:**
    - Rating, search rank, and RPI (Return Product Index) visualization.
    - Configurable threshold-based alerts for performance drops/spikes.
    - Centralized views for category and business teams.

- **Fulfillment Track & Trace Service:**

- Real-time action eligibility engine (address change, cancellation, etc.).
- Efficient storage/retrieval of fulfillment metadata for low-latency decisions.
- Designed reporting layer for SLA adherence and business dashboards.
- Supported two architecture versions for gradual scalability improvements.

Technology Stack

- **Frontend:** React.js, React Native, Progressive Web App (PWA)
- **Backend:** Spring Boot microservices, Node.js (Express middleware), Amazon Redshift
- **Data Stores:** MongoDB (with transaction handling), Elasticsearch
- **Architecture:** Microservices with versioned abstraction layers
- **Delivery Responsibilities:** Requirements gathering, HLD/LLD, data store design, full-stack development, E2E deployment, functional & non-functional testing, alerting/monitoring, and post-production support

Customer visit : more then 80 lac/per day on BBD days

Project 5:

Aeromexico – Digital Flight Booking & Travel Experience Platform

Organization: Aeromexico (via employer)
Role: Senior Web Developer / Backend Engineer **Duration:** (As per resume timeline)

Overview
Built a full-stack B2C airline platform enabling flight search, booking, trip management, check-in, loyalty integration, and post-booking services. The platform supports multi-currency, multilingual experiences and operates reliably at high traffic across international markets.

Key Responsibilities & Contributions

- Developed core booking, pricing, and reservation services.
- Implemented trip management, check-in, ancillaries, and loyalty integration flows.
- Contributed to personalization and offer modules for cross-sell and promotions.

Key Functional Modules

- **Flight Search, Booking & Pricing Engine:**
    - One-way, round-trip, and multi-city search with real-time availability.
    - Branded fares with baggage/flexibility/benefits breakdown.

    - Multi-currency and tax/fee handling with transparent fare display.
- **Trip Management & Check-In:**
    - "My Trips" retrieval and itinerary management.
    - Online check-in, seat selection, boarding pass generation.
    - Changes, cancellations, refunds, flight status, ancillaries.
- **Loyalty & Personalization (Aeromexico Rewards):**
    - Login, account linking, and reward point calculation.
    - Post-booking loyalty updates and personalized cross-sell (hotel, car, insurance).
- **Travel Information & Support:**
    - Structured content for policies and requirements.
    - Multilingual support and SEO-optimized pages.
    - Embedded support entry points (chat/phone/forms).

Technology Stack

- **Backend Services:** Java (Spring Boot) for booking/pricing/reservations, NestJS
- **Middleware & APIs:** Node.js (Express) as API gateway/BFF integrating reservation, loyalty, and payment systems
- **Frontend:** React.js SPA, ES6, Redux/Redux Thunk, responsive web design
- **APIs & Integration:** REST, GraphQL, airline GDS/host systems, loyalty APIs
- **Event Processing:** Kafka/message queues for async workflows (notifications, audit, sync)
- **Security:** JWT-based authentication, role-aware access, PCI-compliant payment flows
- **DevOps:** Dockerized services, CI/CD pipelines, multi-environment cloud deployments

Project 6:

Philips – Connected Healthcare Patient–Practitioner Platform

Organization: Philips (via employer)
Role: Senior Web Developer / Backend Engineer

Overview

Developed a cloud-based healthcare platform facilitating secure, continuous engagement between patients, practitioners, and administrators. Centralized clinical records, remote monitoring, and questionnaire-based assessments using HL7 FHIR (miFhir) standards.

Key Responsibilities & Contributions

- Implemented core backend services for patient data, questionnaires, appointments, and clinical workflows.
- Built practitioner, patient, and admin portals for end-to-end care coordination.
- Designed data models aligned with HL7 FHIR standards and integrated various clinical modalities.

Key Functional Modules

- **Practitioner Portal:**
    - Patient lists, appointment views, disease history, clinical notes.
    - Survey assignment, medication management, diagnostics ordering.
    - Visualization of labs, vitals, ECGs, and measurements.
    - Integrated messaging and questionnaire editor.
- **Patient Portal:**
    - Appointment timeline, disease overview, vitals and lab data entry.
    - Survey completion, medication tracking, Excel export for records.
    - Q&A flow with practitioners.
- **Admin & Configuration Module:**
    - User and role management (patients/practitioners/sub-admins).
    - Disease master data and configuration of activities/observations.
    - Backup and restore console for data integrity.

Technology Stack

- **Backend:** Node.js, Spring Boot microservices, NestJS
- **Healthcare Standards:** HL7 FHIR with miFhir-based data model
- **Database:** MongoDB

- **Frontend (Web):** Angular 4, jQuery, React-animate
- **Build Tools:** Bower, Grunt

- **Containerization & Hosting:** Docker, AWS
- **Mobility:** React + Ionic-based mobile applications

Project 7:

Legato – Workflow Orchestration & Automation Platform

Organization: Seecog Softwares / Product Initiative

Overview
Built a workflow orchestration platform that models business processes as directed graphs, enabling configurable, automated, and repeatable execution of complex routines across services and systems.

Key Responsibilities & Contributions

- Designed the core workflow engine and graph-based modeling concepts.
- Implemented services for workflow definition, execution, monitoring, and administration.
- Integrated external systems via REST APIs and built robust error-handling patterns.

Key Functional Modules

- **Workflow Modeling & Orchestration:**
  - Graph-based workflows with parallel/sequential paths and priority.
  - Versioned scenarios for safe rollout and controlled changes.
- **Task Execution & Service Integration:**
  - Task execution engine with state transitions and IO handling.
  - Sync/async patterns for long-running operations.
  - Retry, timeout, and error-handling strategies.
- **Monitoring, Logging & Administration:**
  - Web console for workflow runs, logs, and statuses.
  - Reporting on execution history, success/failure, performance.
  - Administrative controls (pause, resume, cancel, reconfigure).

Technology Stack

- **Backend:** Golang-based workflow engine and APIs, Spring Boot microservices, NestJS
- **APIs:** RESTful APIs for defining, triggering, and querying workflows
- **Web Application:** Browser-based console for configuration and monitoring
- **Testing:** Website, API, and integration testing for end-to-end validation
- **Architecture:** Service-oriented design focused on reliability, fault tolerance, and extensibility

Project 8:

Fleetzz – AI-Powered Fleet Tracking & Anti-Theft Platform

Organization: Fleetzz (Product)

Overview

Engineered an AIS-140–compliant, AI-powered fleet tracking and anti-theft platform for vehicle owners, transporters, emergency services, and enterprises. The platform combines GPS hardware with cloud software for real-time visibility, route optimization, and security automation.

Key Responsibilities & Contributions

- Designed polyglot microservices for tracking, alerts, trip analytics, and user management.
- Helped define IoT-to-cloud data flows for time-series tracking and event processing.
- Contributed to dashboards for real-time control tower views, alerts, and reporting.

Key Functional Modules

- **Real-Time Tracking & Fleet Visibility:**
  - Live map tracking, trip playback, idle/deviation analysis.
  - Role-based dashboards for individual owners, fleets, and enterprises.
- **Anti-Theft & Security Automation:**
  - Movement alerts for unauthorized motion.
  - AI-based anomaly detection on routes, stops, and patterns.
  - E-sealing and tamper detection with centralized alerting.

- **Route Optimization & Transport Operations:**
    - Automated tracking workflows and ETA visibility.

    - Route planning and performance-based optimization.
    - Operational reports for billing, SLAs, and cost control.
- **Emergency Services & Smart Response:**
    - Specialized workflows for ambulances/emergency vehicles.
    - Traffic-aware routing, geofencing around hospitals, and response-time analytics.
- **Enterprise Integration & Compliance:**
    - APIs for TMS/ERP/WMS integration.
    - Multi-fleet/multi-region operations with RBAC and business rules.
    - AIS-140 compliance via certified devices.

Technology Stack

- **Device & IoT Layer:** AIS-140–compliant GPS devices with movement/ignition/tamper sensing
- **Backend Services:** Polyglot microservices using Java (Spring Boot) for core services; Python (FastAPI/Django) for data processing and background jobs; NestJS for additional services
- **Data & Storage:** Time-series + relational storage for locations, trips, geofences, alerts; analytics layer for reporting/anomaly detection
- **Frontend (Web):** Web dashboards for owners, fleet managers, and admins with live maps and alert consoles
- **Mobile Apps:** Android/iOS apps for tracking and alert management
- **Mapping & Routing:** Map SDKs and routing APIs for live location, traffic-aware routing, ETA, and geofencing
- **Security & Notifications:** RBAC, secure device-to-cloud communication, multi-channel notifications (in-app, SMS, email/push)